Ch 6.1: Subset Selection

Lecture 16 - CMSE 381

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Announcements

Last time

- k-fold CV for Classification

Covered in this lecture

- Subset selection
- Forward and Backward Selection

Announcements:

- HW #4 Due Tonight!

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Subset selection	6.1	HW #4 Due Weds 10/9
17	Fri	10/11	Shrinkage: Ridge	6.2.1	
18	Mon	10/14	Shrinkage: Lasso	6.2.2	
19	Wed	10/16	Dimension Reduction	6.3	
20	Fri	10/18	Overflow, Possibly more dimension reduction?		HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		

Section 1

Last time

Goals of fitting a given model

Up to now, we've focused on standard linear model: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$ and done least squares estimation. *Translation: Minimize RSS* $= \sum_i (y_i - \hat{y}_i)^2$

Prediction accuracy

- If the true relationship is approximately linear, least squares estimates have low bias
- If $n \gg p$, then least squares also has low variance
- If n not much larger than p, then high variability, overfitting, poor predictions
- If $n < p$ then no unique solution so can't use at all
- Goal on next slide: shrink/constrain number of variables to improve accuracy

Goals of fitting a given model

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Model Interpretability

- Often, many variables included aren't associated with response
- Including them leads to unnecessary complexity in the model
- Idea: Set these coefficients to 0 (or close to zero)
- Goal on next slide: Do some automatic feature selection / variable selection to get rid of unnecessary variables

Section 2

Best Subset Selection

Too many variables

All subsets of 4 variables ($2^4 = 16$)

- $\emptyset$
- X_1
- X_2
- X_3
- X_4
- $X_1\ X_2$
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The game: fit the model 2^n times, score each, pick one with best score

One way of breaking this up

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

- Part 2b goes for lowest training score, Part 3 then goes for lowest testing score.
- Go back to previous page and walk through the algorithm and variables slowly
- point out that Step 2 is computational infeasable for large p

Calculate by hand

We train a model using four variables, X_1, X_2, X_3, X_4. We're interested in getting a subset of the variables to use. The following table shows the mean squared error and the MSE value computed for the model learned using each possible subset of variables.

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
X2	7.42	8.01
X3	8.16	8.3
X4	8.33	9.06
X1,X2	4.33	7.47
X1,X3	5.82	5.22
X1,X4	3.17	4.23
X2,X3	4.07	3.78
X2,X4	3.31	4.01
X3,X4	3.06	4.16
X1,X2,X3	3.08	5.49
X1,X2,X4	3.55	4.02
X1,X3,X4	2.97	4.23
X2,X3,X4	2.98	3.17
X1,X2,X3,X4	2.16	4.39

1. What subset of variables is found for each of the sets $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ when using best subset selection?
2. What subset of variables is returned using best subset selection?

Extra work space if it helps

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$\mathcal{M}_0 = \emptyset, \mathcal{M}_1 = \{X_2\}, \mathcal{M}_2 = \{X_3, X_4\}, \mathcal{M}_3 = \{X_1, X_3, X_4\}, \mathcal{M}_4 = \{X_1, X_2, X_3, X_4\}, .$

Among test error, best is $X_3 X_4$

Code to do this

Section 3

Forward Selection

What's the problem?

- Checking 2^p models is not reasonable for large p, $p > 40$
- The next bits are finding alternatives to Step 2

Forward Stepwise Selection

Algorithm 6.2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
2. For $k = 0, \ldots, p-1$:
   (a) Consider all $p-k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
   (b) Choose the *best* among these $p-k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

An example for Forward Stepwise Selection

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- X_3
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Group work: by hand same example with forward example

We train a model using four variables, X_1, X_2, X_3, X_4. We're interested in getting a subset of the variables to use. The following table shows the mean squared error and the R^2 value computed for the model learned using each possible subset of variables.

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1. What subset of variables is found for each of the sets $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ when using forward selection?
2. What subset of variables is returned using forward subset selection?

Extra work space if it helps

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Among test error, best is $X_3 X_4$

Pros and Cons of Forward Stepwise

Pros:

- Computationally cheaper
- Number of models fit is

$$1 + \sum_{k=0}^{p-1}(p-k) = 1 + \frac{p(p+1)}{2}$$

which is way better than 2^p

Cons:

- Not guaranteed to find the best model
- As example: if best 1-variable model is X_1, but best 2-variable model is X_2X_3, then forward selection won't find it.
- Is this a con? Maybe just a limitation If $n < p$, then can only construct models $\mathcal{M}_0 \cdots \mathcal{M}_{n-1}$

Section 4

Backward Selection

Backward stepwise selection

Algorithm 6.3 *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
2. For $k = p, p-1, \ldots, 1$:
   (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k-1$ predictors.
   (b) Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

Pros and Cons of Backward Stepwise

Pros:

- Computationally cheaper
- Number of models fit is still

$$1 + \sum_{k=0}^{p-1}(p-k) = 1 + \frac{p(p+1)}{2}$$

which is way better than 2^p

Cons:

- Not guaranteed to find the best model
- Unlike forward selection, this can't be used at all if $n < p$

Can also look at hybrid approaches with both forward and backward selection

TL;DR

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
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3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

- Modify step 2 with forward or backward selection
- Choose best model in step 3 using one of our adjusted training scores or CV

Next time

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